



HK IMO Training 2022-2023
Problem Set 8 Suggested Solutions



Problem 29 (Rioplante MO 2009 Problem 1). Find all real numbers a and b such that whenever both quadratic equations $x^2 + ax + 1 = c$ and $x^2 + bx + 1 = d$ are solvable in real numbers (where c and d are real numbers), the quadratic equation $x^2 + (a+b)x + 1 = cd$ must be solvable in real numbers.

Solution. Equivalently, we need to find all real numbers a and b such that whenever

$$a^2 - 4(1 - c) \geq 0, \quad (1)$$

$$b^2 - 4(1 - d) \geq 0, \quad (2)$$

then we must have

$$(a + b)^2 - 4(1 - cd) \geq 0. \quad (3)$$

The conditions (1) and (2) can be rewritten as

$$c \geq \frac{4 - a^2}{4}, \quad (4)$$

$$d \geq \frac{4 - b^2}{4}. \quad (5)$$

If $a^2 > 4$, we can choose c to be a negative number and choose d to be a sufficiently large number. Then cd is a sufficiently small number, so that (3) cannot hold. Therefore, we must have $-2 \leq a \leq 2$. Similarly, $-2 \leq b \leq 2$.

Next, consider $c = \frac{4 - a^2}{4}$ and $d = \frac{4 - b^2}{4}$. In that case, (3) is the same as

$$(a + b)^2 - 4 \left(1 - \frac{(4 - a^2)(4 - b^2)}{16} \right) \geq 0,$$

which means

$$ab(ab + 8) \geq 0.$$

In order that this holds, we must have $ab \geq 0$ since $ab + 8 \geq 8 - |ab| \geq 8 - (2)(2) > 0$. In other words, we have $0 \leq a, b \leq 2$ or $-2 \leq a, b \leq 0$.

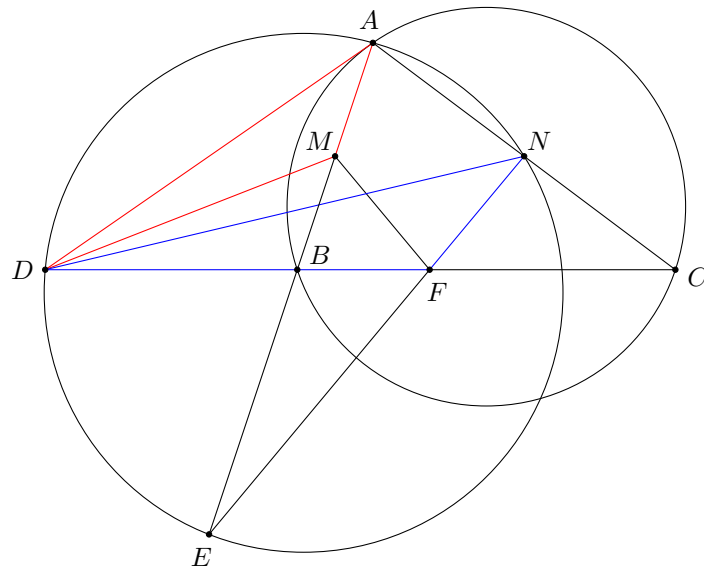
Lastly, we claim that all pairs a, b satisfying $0 \leq a, b \leq 2$ or $-2 \leq a, b \leq 0$ are solutions. Indeed, whenever (4) and (5) hold, we must have $c, d \geq 0$. Therefore,

$$(a + b)^2 - 4(1 - cd) \geq (a + b)^2 - 4 \left(1 - \frac{(4 - a^2)(4 - b^2)}{16} \right) = ab(ab + 8) \geq 0.$$

This completes the proof. □

Problem 30. In $\triangle ABC$, $AB < AC$. Let M and N be the midpoints of AB and AC respectively. Let D be the intersection point of the line BC and the tangent at A to the circumcircle of $\triangle ABC$. The circumcircle of $\triangle ADN$ meets the line AB again at E . The lines BC and EN meet at F . Prove that DM is tangent to the circumcircle of $\triangle BFM$.

Solution. Note that $\triangle DAB \sim \triangle DCA$. As M and N are the midpoints of AB and CA , we have $DAMB \sim DCNA$. Firstly, this implies $\angle ADM = \angle CDN = \angle NDF$. Together with $\angle DAM = \angle DNF$, we obtain $\triangle DAM \sim \triangle DNF$. By spiral similarity, this yields $\triangle DAN \sim \triangle DMF$.



Secondly, we have $\triangle DAN \sim \triangle DBM$ from the above similarity. It follows that $\triangle DMF \sim \triangle DBM$. Therefore, DM is tangent to (BFM) as desired. $\square$

Problem 31 (Bulgaria MO 2015 Problem 2). Find the smallest positive integer n that satisfies the following condition: It is possible to choose exactly 101 cells of an $n \times n$ table, such that there is a unique way to partition the table into rectangles (by cutting along the grid lines of the table) where each rectangle contains exactly one chosen cell.

Solution. The answer is 101. Firstly, when $n = 101$, we can choose all cells in the first row. It is clear that the only partition is to divide the table into 101 columns, so it satisfies the condition.

We now show that $n \geq 101$. We first provide a way to form an admissible partition. Initially, we draw 101 rectangles of size 1×1 which are exactly the 101 chosen cells. Then we perform the following 4 operations in any order.

- (1) We extend all rectangles to the left to include as many cells as possible (of course the rectangles should not overlap) until this cannot be done. Note that the result does not depend on the order of the rectangles that we select.
- (2) We extend all rectangles to the right to include as many cells as possible.
- (3) We extend all rectangles upwards to include as many cells as possible.
- (4) We extend all rectangles downwards to include as many cells as possible.

After these operations, we must obtain a partition of the table since all cells must be included. Also, each of the 101 rectangles contains exactly one chosen cell. The condition asserts that no matter which order we choose, the resultant partition is the same.

Firstly, observe that all chosen cells in the same row (or the same column) must be contiguous. Indeed, if there is at least one non-chosen cell between two chosen cells in the same row (assuming there are no other chosen cells in between), then the operations (1)(2)(3)(4) and (2)(1)(3)(4) will generate different partitions as the cells between these two cells belong to different rectangles in these two partitions. This is a contradiction.

Secondly, if (i, j) and (k, ℓ) are chosen cells where $i \neq k$ and $j \neq \ell$ (where i, k are the row indices counted from the top, while j, ℓ are the column indices counted from the left), then (i, ℓ) and (k, j) must be chosen cells. To prove this, WLOG assume $i < k$ and $j < \ell$. If the cell (i, ℓ) is not chosen, then the operations (2)(1)(3)(4) and (3)(4)(2)(1) will generate different partitions as this cell will belong to different rectangles, contradiction. Therefore, (i, ℓ) must be chosen. Similarly, (k, j) must be chosen.

It follows from the above two observations that all chosen cells form a rectangle. As 101 is a prime, all chosen cells must form a 1×101 or a 101×1 rectangle. In either case, we need $n \geq 101$. This proves the assertion. $\square$

Problem 32 (Balkan MO 2018 Problem 4). Find all prime numbers p and q such that $3p^{q-1} + 1$ divides $11^p + 17^p$.

Solution. Consider any odd prime r of $3p^{q-1} + 1$. Then $r \mid 11^p + 17^p$. This implies $r \neq 11, 17$. Let d be the order of $11 \cdot 17^{-1}$ modulo r . As $(11 \cdot 17^{-1})^p \equiv -1 \pmod{r}$, we have $(11 \cdot 17^{-1})^{2p} \equiv 1 \pmod{r}$, and hence $d \mid 2p$. Since p is a prime number, we have $d = 1, 2, p, 2p$. The cases $d = 1, p$ are impossible since $(11 \cdot 17^{-1})^p \equiv -1 \not\equiv 1 \pmod{r}$.

- If $d = 2$, then $(11 \cdot 17^{-1})^2 \equiv 1 \pmod{r}$. This implies $r \mid 289 - 121 = 168$, and thus $r = 3, 7$. The case $r = 3$ is impossible since $3 \nmid 3p^{q-1} + 1$.
- If $d = 2p$, then $2p \mid r - 1$ by the property of order. In particular, we have $r \equiv 1 \pmod{p}$.

We first handle the special case $p = 2$. We need to solve $3 \cdot 2^{q-1} + 1 \mid 11^2 + 17^2 = 410$. When $q = 2, 3, 5, 7$, we have $3 \cdot 2^{q-1} + 1 = 7, 13, 49, 193$, none of which divides 410. When $q \geq 11$, we have $3 \cdot 2^{q-1} + 1 > 410$.

Now, suppose p is an odd prime. By the lifting the exponent lemma, we have $v_7(11^p + 17^p) = v_7(11 + 17) + v_7(p) = 1 + v_7(p)$. If $7 \mid p$, then $7 \nmid 3p^{q-1} + 1$. If $7 \nmid p$, then $7 \parallel 11^p + 17^p$, and so $v_7(3p^{q-1} + 1) = 0, 1$. In any case, we have $v_7(3p^{q-1} + 1) = 0, 1$.

If q is an odd prime, then $q - 1$ is even, and hence $3p^{q-1} + 1 \equiv 3 + 1 = 4 \pmod{8}$. This implies $3p^{q-1} + 1 = 4r_1 \cdots r_k$ or $28r_1 \cdots r_k$ for some (possibly equal) odd primes r_j satisfying $r_j \equiv 1 \pmod{p}$. Considering modulo p , we obtain $1 \equiv 4 \pmod{p}$ or $1 \equiv 28 \pmod{p}$. In either case, the only possibility is $p = 3$. This gives

$$11^p + 17^p = (11 + 17)(11^2 - 11 \cdot 17 + 17^2) = 2^2 \cdot 7 \cdot 223.$$

- When $q = 3$, $3p^{q-1} + 1 = 28$ divides $11^p + 17^p$.
- When $q = 5$, we have $3p^{q-1} + 1 = 244 \nmid 11^p + 17^p$.
- When $q = 7$, we have $3p^{q-1} + 1 = 2188 \nmid 11^p + 17^p$.
- When $q \geq 11$, we have $3p^{q-1} + 1 > 100000 > 11^p + 17^p$.

If $q = 2$, we need to solve $3p + 1 \mid 11^p + 17^p$. As $11^p + 17^p \equiv 11 + 17 \equiv 4 \pmod{8}$, we can write $3p + 1 = 2r_1 \cdots r_k$ or $4r_1 \cdots r_k$ for some (possibly equal) odd primes r_j satisfying $r_j \equiv 1 \pmod{p}$. This implies $1 \equiv 2 \pmod{p}$ or $1 \equiv 4 \pmod{p}$. The only possibility is $p = 3$. But then $3p + 1 = 10 \nmid 11^p + 17^p$.

To conclude, the only solution is $p = q = 3$. □